

DaNae R. Woodard, Ph.D.
San Diego IRACDA Postdoctoral Fellow
San Diego, CA
woodard.danae@gmail.com

EDUCATION

University of Texas Southwestern Medical Center, Dallas, TX Aug 2015 – Dec 2021
Graduate School of Biomedical Science
Doctor of Philosophy in Biological Chemistry

University of Houston-Downtown, Houston, TX Aug 2010 – May 2015
Bachelor of Science in Chemistry

TEACHING EXPERIENCE

Cell and Molecular Biology (BIOL366) Aug 2024 – present
Department of Biology
University of California, San Diego

Lecturer

STARTneuro Program Aug 2024

Introduction to Molecular Neuroscience, Genes and Gene Manipulation lecture

Neurobiology of the Visual System lecture

University of California, San Diego

Guest Lecturer

Human Anatomy (BIOL 212) Mar 2024

Brain and Spinal Cord lectures

San Diego State University

Guest Lecturer

Introduction to Cell and Molecular Biology (BIOL 210A) Nov 2023

Energy Metabolism lecture

San Diego City College

Guest Lecturer

SD IRACDA National Institute on Scientific Teaching Workshop Nov 2023 – Dec 2023

Exploring how student experiences shape learning

Hosted by SD IRACDA

University of California, San Diego

San Diego City College

San Diego State University

Co-leader

Human Genetics and Genomics Course Mar 2022 – present

University of California, San Diego

Teaching Assistant

Introduction to College Teaching Course May 2022 – June 2022

University of California, San Diego

D.R. Woodard

Active Participant

RESEARCH EXPERIENCE

University of California, San Diego, San Diego, CA Feb 2022 – Present
Principle Investigator: Dr. Radha Ayyagari
Department of Ophthalmology
SD IRACDA (K12) Fellow
Postdoctoral Fellow

University of Texas Southwestern Medical Center, Dallas, TX Dec 2018 – Dec 2021
Principle Investigator: Dr. John D. Hulleman
Department of Ophthalmology
Biological Chemistry Program
Graduate Student

University of Texas Southwestern Medical Center, Dallas, TX Mar 2016 – Nov 2018
Principle Investigator: Dr. Marc I. Diamond
Center for Alzheimer's and Neurodegenerative Diseases
Biological Chemistry Program
Graduate Student

Texas A&M University, College Station, TX June 2014 – Aug 2014
Principle Investigator: Dr. Margaret E. Glasner June 2013 – Aug 2013
Department of Biochemistry and Biophysics
Summer undergraduate researcher

PUBLICATIONS

11. Biswas, P*, **Woodard DR***, Hollingsworth, TJ., Khan, N., Lazaro, D, Berry, AM., Dagar, M., Pan, Y., Garland, D., Shaw PX., Oka, C., Iwata, T., Jablonski, M., Ayyagari, R. Ablation of Htra1 leads to the development of sub-RPE deposits and photoreceptor abnormalities in mice. (*- equal contribution). Submitted to *Journal of Clinical Insights*. (2023). Submitted. In revision.
10. Hayes, HM., **Woodard DR.**, Hulleman JD. (2023). Ocular Amyloid, Condensates, and Aggregates-Higher-Order Protein Assemblies Participate in Both Retinal Degeneration and Function. *Adv Exp Med Biol*. 2023; 1415:263-267. PMID: 37440043.
9. **Woodard DR.**, Daniel S., Nakahara E., Abbas A., DiCesare SM., Collier GE., Hulleman JD. A loss-of-function cysteine mutant in fibulin-3 (EFEMP1) forms aberrant extracellular disulfide-linked homodimers and alters extracellular matrix composition. *Hum Mutat* (2022). PMID: 33542268.
8. Peng, H.*, Ramadurgum, P.*, **Woodard, DR***, Daniel, S, Renwick, M., Datta, S., Chen, B., Aredo, B., Ufret-Vincenty, RL., Hulleman, J.D. (2021). Utility of the DHFR-based destabilizing domain across mouse models of retinal degeneration and aging. (*- equal contribution). *iScience*. (2022). May 20; 25(5): 104206, PMID: 35521529.
7. Manna, MS., Tamer, YT., Gaszek, I., Poulides, N., Ahmed, A., Wang, X., Toprak, FCR., **Woodard, DR.**, Koh, A.Y., Williams, N., Borek, D., Atilgan, A.R., Hulleman, JD., Atilgan, C., Tambar, U.K., and Toprak, E. A trimethoprim derivative impedes antibiotic resistance evolution. *Nat. Commun.* (2021) May 19;12(1):2949. PMID: 34011959.

6. **Woodard DR.**, Xing C., Ganne P., Liang H., Mahindrakar A., Sankurathri C., Hulleman JD[#], and Mootha VV[#]. A novel homozygous P388S mutation in TULP1 causes protein instability and retinitis pigmentosa. *Mol. Vis.* (2021); 27:179-190. ([#] - co-corresponding author). PMID: 33907372.
5. **Woodard DR.**, Nakahara E., Hulleman JD. Clinically-identified C-terminal mutations in fibulin-3 are prone to misfolding and destabilization. *Sci. Rep.* (2021) Feb 4;11(1):2998. PMID: 33542268.
4. Ramadurgum P., **Woodard DR.**, Daniel S., Peng H., Mallipeddi PL., Niederstrasser H., Mihelakis M., Chau VQ., Douglas PM., Posner BA., Hulleman JD. Simultaneous Control of Endogenous and User-Defined Genetic Pathways Using Unique ecDHFR Pharmacological Chaperones. *Cell Chem. Biol.* (2020) May 21;27(5):622-634.e6. PMID: 32330442.
3. Chen D., Drombosky KW., Hou Z., Sari L., Kashmer OM., Ryder BD., Perez VA., **Woodard DR.**, Lin MM., Diamond MI., Joachimiak LA. Tau local structure shields an amyloid-forming motif and controls aggregation propensity. *Nat. Commun.* (2019) Jun 7;10(1):2493. PMID: 31175300.
2. Sharma AM., Thomas TL., **Woodard DR.**, Kashmer OM., Diamond MI. Tau Monomer Encodes Strains. *eLife*: (2018); 7: e37813. PMID: 30526844.
1. Odokonyero D., Ragumani S., Lopez MS., Bonanno JB., Ozerova ND., **Woodard DR.**, Machala BW., Swaminathan S., Burley SK., Almo SC., Glasner ME. Divergent Evolution of Ligand Binding in the o-Succinylbenzoate Synthase Family. *Biochemistry* (2013) 52: 751221. PMID: 24060347.

MEMBERSHIPS

NIH/NIGMS Institutional Research and Career Development Award (IRACDA) Sept 2022 – Present
San Diego IRACDA (SD IRACDA) Fellow

The Association for Research in Vision and Ophthalmology (ARVO) Postgraduate Member-in-Training status	April 2022 – Present
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Biochemistry Department Diversity and Inclusion Committee University of Texas Southwestern Medical Center Graduate Student Representative	Sept 2020 – Dec 2021
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The Association for African American Scientists (TAAAS) University of Texas Southwestern Medical Center Speaker Committee Chair	Jan 2017 – Jan 2019
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RESEARCH CONFERENCES

The Association for Research in Vision and Ophthalmology (ARVO) Seattle, WA (2024) New Orleans, LA (2023)	April 2023, May 2024
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RD2023 International Symposium of Retinal Degeneration Meeting Torremolinos, Spain	Oct 2023
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The National IRACDA Conference San Antonio, TX (2023) Chapel Hill, NC (2024)	June 2023, June 2024
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Annual Biomedical Research Conference for Minority Students (ABRCMS) Nov 2020
Virtual

Biomedical Science Careers Student Conference April 2018
Boston, MA

RESEARCH PRESENTATIONS

ARVO May 2024
Multi-omic analysis of RPE and photoreceptor pathology in early-onset retinal degeneration in Mfrp^{-/-} mice
Poster

RD2023 International Symposium of Retinal Degeneration Meeting Oct 2023
Integrative single-nucleus multi-omic analysis of retinal cell types involved in early-onset retinal degeneration due to the loss of Mfrp
RD2023 Travel Fellowship Awardee
Poster

SD IRACDA SURF Program Laboratory Research Presentation July 2023
Invited Speaker

National IRACDA Conference June 2023
Single-Nucleus Multi-modal Analysis of Photoreceptor Alterations during Cellular Stress in MFRP-associated Retinal Degeneration
Poster

ARVO April 2023
Single-Nucleus Multi-modal Analysis of Photoreceptor Alterations during Cellular Stress in MFRP-associated Retinal Degeneration
ARVO Foundation Travel Grant Awardee
Poster

NIH Bridge2AI AI Ready and Equitable Atlas for Diabetes Insights Meeting Jan 2023
University of California, San Diego
Panel Speaker

Women in Science Seminar Nov 2022
University of Houston-Downtown
Keynote Speaker

SD IRACDA SURF Program Professionalism and Research Ethics Presentation July 2022
Invited Speaker

HONORS/AWARDS

NanoString Technologies WTA Neuroscience Grant Program Awardee Mar 2023

SD IRACDA Best Speaker/Presentation Awardee May 2023

NIH/NIGMS SD IRACDA K12 Awardee	Sept 2022 – Present
National Institutes of Health Diversity Research Supplement (R01 EY027785) Hulleman Lab	July 2020 – Jan 2022
National Institutes of Health Diversity Research Supplement (R01 AG048678) Diamond Lab	Aug 2018 – April 2019
University of Houston-Downtown Student Commencement Speaker	May 2015

OUTREACH

UCSD Summer Research Conference Assisted with directing guests to student presentations and posters	Aug 2024
Barrio Logan Science & Art Expo Science demonstration booths setup volunteer	April 2024
San Diego State University IMSD/MARC Mock Interviews Conducted mock graduate school interviews for undergraduates	Jan 2023 Jan 2024
Dallas County Community College Presented scientific research to undergraduates	June 2021
Harmony Science Academy Carrollton Science and Technology Fair Judged middle school biology and chemistry projects	Jan 2020
UT Southwestern Graduate School Recruiting Student Host Highlighting graduate school to interview candidates	Jan 2017 – Dec 2021
UT Southwestern STARS Summer Research Program Assist STARS high school students with lab projects	June 2018 June 2019
UT Southwestern Science in the City Conducted lab tours and science stations for public attendees	May 2018
Foundation for C.H.O.I.C.E. High School Student Mentor Encouraged at-risk students to pursue STEM college/career opportunities	Aug 2016 – Aug 2018